

Dr Jerome Graves

R&D Engineer | Ultrasonics, Instrumentation & Robotics | PhD

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Professional Profile

Hands-on R&D engineer (PhD in **ultrasonic imaging**, 1st-class BEng) who designs, builds, and validates **integrated hardware–software systems**. Deep background in **ultrasonics, acoustic wave modelling, and signal processing**, applied through the design and build of custom experimental systems, including an **automated multi-axis ultrasonic scanning system** (mechanical design, acquisition electronics, **C/C++** firmware, motion control, analysis software) and a camera-based inspection instrument with a **machine-vision** pipeline.

Technical lead on a **£100k industrial automation project** (2× productivity gain), with **two papers accepted at the IEEE International Ultrasonics Symposium 2026**. Comfortable owning the full development lifecycle: concept, prototype construction, **experiment design and validation**, data analysis, and client-facing delivery.

Technical Skills

Prototyping & Integration: experimental rig, fixture, and test-hardware design; mechanical CAD (Fusion 360) and 3D printing; PCB design in KiCad with DFM for volume manufacture; assembly, bring-up, and hardware–software debug.

Acoustics, Imaging & Signal Processing: ultrasonic transducers and instrumentation, acoustic wave propagation modelling, DSP, imaging and inversion algorithms, machine vision (OpenCV: camera calibration, homography rectification, segmentation), validation against experiment.

Software: C++ (Visual Studio), Python, MATLAB, scripting for automation, testing, and data analysis; Git, CMake, Bash/Linux, parallel computing, LaTeX.

Robotics, Embedded & Control: motion control, ROS, sensing and actuation, C/C++ firmware on microcontrollers and FreeRTOS (ESP-IDF), real-time systems, FPGA/VHDL (Quartus Prime, ModelSim).

Experimentation & Validation: experiment design, automated data acquisition, throughput and reliability testing, data analysis and interpretation, technical reporting and documentation.

Key Achievements

- Designed and built an **automated multi-axis ultrasonic scanning system** from scratch: mechanical design, acquisition electronics, embedded firmware, motion control, and analysis software integrated into a single validated platform.
- Led technical delivery on a **£100,000 KTP project**, contributing to a redesigned industrial system that achieved a **2× productivity gain**.
- Led a **£19,000 industry-funded ultrasound consultancy**, designing experiments and delivering client-facing validation and reports.
- Authored **two papers accepted at the IEEE International Ultrasonics Symposium 2026**, with a further journal article under review.
- Built **parallel-processing analysis pipelines** for large ultrasonic datasets, cutting analysis runtime by **70%**.
- Led **LSBU's UniBots UK robotics team** to **2nd place (2024)** and **3rd place (2023)**, managing teams of up to seven students.

Education

PhD London South Bank University, Electronic Engineering

Sept 2022 – July 2026

- Title:** Three-Dimensional Ultrasound Imaging of the Crystal Structure of Ice Cores.
- Designed and executed **ultrasonic experiments** end to end: transducer selection and instrumentation, acquisition, signal processing, and validation against numerical models.

- Built an **automated multi-axis ultrasonic scanning rig** integrating mechanical, electronic, firmware, and control-software subsystems.
- Developed **acoustic wave propagation simulation** and inversion tools in MATLAB and Python for anisotropic media.

BEng London South Bank University, Electrical and Electronic Engineering (1st Class)

Sept 2017 – June 2020

- **Final Year Project:** Designed a ROS-based quadrupedal robot with embedded control, mechanical design, and terrain-adaptive motion.
- **Awards & leadership:** School of Engineering Award for Excellence in Student Creativity (peer-voted, one of four recipients); led a team to the **National Finals** of the Engineering for People Design Challenge (Engineers Without Borders UK).

Professional Experience

London South Bank University, Doctoral Researcher (SPE3D Ultrasonic Lab)

London, UK

Sept 2022 – July 2026

- Designed and built an **automated multi-axis ultrasonic scanning system:** embedded acquisition firmware on **microcontrollers**, **motion control**, and integrated mechanical, electronic, and control-software subsystems.
- Built a **camera-based optical inspection instrument:** three-axis motion control on a custom PCB, polarised LED lighting, and a Python/OpenCV **machine-vision pipeline** for camera calibration, image rectification, and grain segmentation.
- Designed and executed experiments characterising **ultrasonic wave propagation in anisotropic media**, from rig setup and data acquisition through analysis and interpretation.
- Developed automated analysis and simulation pipelines in **MATLAB and Python**, cutting analysis runtime by **70%** through parallel processing.
- Validated experimental results against numerical models, feeding findings back into apparatus design and measurement methodology.

London South Bank University, Lecturer / Teaching Assistant

London, UK

Jan 2023 – 2026

- Delivered teaching and practical support in **Python, MATLAB, instrumentation, and engineering systems**.
- Supported student projects involving software development, embedded systems, and electromechanical integration.
- Developed technical teaching resources and assessed project and laboratory work.

London South Bank University & KURE, Research Lead in Ultrasound Testing

London, UK

2022 – 2023

- Led a **£19,000 industry-funded consultancy** in ultrasound testing.
- Designed and executed automated experimental workflows and technical analysis for **client-facing product development**.
- Delivered validation reports used directly by the client to support engineering and product decisions.

London South Bank University & Industrial Partner, Technical Lead (Knowledge Transfer Partnership)

London, UK

Jan 2021 – Jan 2022

- Led technical delivery on a **£100,000 Knowledge Transfer Partnership** between LSBU and an industrial partner.
- Developed **FPGA/SoC motion-control functionality in VHDL** for an industrial electromechanical platform, with **HDL simulation in Quartus Prime and ModelSim**.
- Led **verification and timing/logic debug** of embedded control and **motion-coordination** behaviour, contributing to a **2× productivity gain** in the re-designed system.
- Delivered documentation, prototypes, and system-level designs feeding into

deployment and manufacture.

Selected Projects & Open-Source Work

OpenUSCT [↗](#) – Ultrasound Computed Tomography Platform

Open source

- Open-source **full-waveform inversion** platform: Python core with **C++ and GPU backends**, an **FPGA acquisition datapath** co-simulated in SystemVerilog, and a browser-based studio interface.

Ultrasonic Crystal-Orientation Estimation and Inversion

PhD

- Estimated crystal orientation from **azimuthal time-of-flight** using an anisotropic (transversely-isotropic) forward velocity model, formulated as a **regularised least-squares inversion** solved by trust-region optimisation.
- Derived a **Cramér–Rao lower bound** on orientation accuracy ($1.7\text{--}2.2^\circ$), using it for forward-model selection and to quantify the measurement's fundamental resolution limit; validated experimentally to a best case of **4.3°** .

Fabric Analyser [↗](#) – Machine-Vision Instrument

Open source

- Automated **cross-polarised imaging instrument**: three-axis stepper gimbal (custom **KiCad** PCB, **C++** firmware) and a **Python/OpenCV** pipeline (calibration, homography, per-grain segmentation) across 72-position scans; a co-registered optical reference validating ultrasonic measurements.

Five-Axis Ultrasonic Core Scanner [↗](#)

Open source

- Pitch-catch transmission scanner for intact ice cores: twin steerable transducer carriages, position-verified Marlin motion, a load-cell coupling interlock, and a Python acquisition stack.

Yertle [↗](#) – 3D-Printed Quadruped Robot (138 GitHub stars)

Open source

- Open-source **ROS-based** legged-locomotion platform: embedded control, mechanical design, and terrain-adaptive motion.

Publications

A Finite-Difference Simulation Framework for Ultrasonic Crystal Orientation Fabric Estimation in Ice: Timing Methodology, Forward Model Selection, and the Cramér–Rao Accuracy Limit, *IEEE International Ultrasonics Symposium*

2026 (accepted)

Jerome Graves, Dr Sevan Harput, Dr Ben Lishman

Non-Destructive Ultrasonic Estimation of Ice Crystal Orientation Fabric: Multi-Frequency Experimental Validation and Failure Mode Characterisation, *IEEE International Ultrasonics Symposium*

2026 (accepted)

Jerome Graves, Dr Sevan Harput, Dr Ben Lishman

The Coding Industry: A Technological Retrospective and Forecast, white paper, [doi:10.5281/zenodo.20887829](https://doi.org/10.5281/zenodo.20887829) [↗](#)

2026

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Measuring Predominant Orientations of Ice Crystal Fabrics From Ultrasonic Measurements of Ice Cores, *Cold Regions Science and Technology*

(preprint)

Jerome Graves, Dr Ben Lishman, Dr Sevan Harput

Determining the Grain Geometry From Ultrasonic Measurements of Large-Grained Temperate Ice Cores, *IEEE International Ultrasonics Symposium*, [doi:10.1109/ius51837.2023.10307539](https://doi.org/10.1109/ius51837.2023.10307539) [↗](#)

2023

Jerome Graves, Dr Ben Lishman, Dr Sevan Harput

An Algorithm for Determining the Grain Geometry From Ultrasonic Measurements of Large-Grained Temperate Ice Cores, *Port and Ocean Engineering under Arctic Conditions*

2023

Jerome Graves, Dr Ben Lishman, Dr Sevan Harput